

The Emergent Spacetime Rendering Engine:

A Systems-Theoretic Interpretation of Holographic Projection, Planck-Scale Quantization,
and Quantum Confinement (v1.0)

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Abstract

Background: Macroscopic three-dimensional spacetime, continuous motion, and rest mass are conventionally treated as fundamental physical entities. This paper advances a systems-theoretic interpretation in which macroscopic physical phenomena emerge as computational artifacts of a discrete, sequential update loop operating on a lower-dimensional holographic boundary.

Formal Core Model: We formalize the discrete boundary evolution as $B_{n+1} = F(B_n)$ and define a phenomenological projection operator $\Pi(B_n) = (g_{\mu\nu}, \Psi, A_\mu, \dots)$ via Multiscale Entanglement Renormalization Ansatz (MERA) tensor network contractions featuring disentanglers u , isometries w , and bond dimension χ with entanglement bound $S \leq \ln \chi$. 3D rest mass is reframed as a projection overhead cost functional $C[B_n]$, dimensionally mapped to local entanglement entropy S_{ent} and tensor-network bond-dimension density, yielding an accounting identity $m \sim \frac{1}{c^2} \frac{\delta C}{\delta \Pi}$.

Falsifiability & Experimental Signatures: We outline four testable experimental signatures: Planck-scale cosmic ray dispersion, holographic interferometer noise, sub-millimeter gravity corrections, and entanglement complexity bounds.

Epistemic Status: We strictly separate heuristic utility from ontological claim, offering the framework as an exploratory research program rather than a finished theory.

1 Introduction & Epistemological Latency of Sensory Reality

Human perception operates under the illusion of a zero-latency, continuous present. However, physical observation is inherently delayed by finite signal propagation speeds and biological processing limits.

1.1 Epistemic Status and Methodology

To maintain rigorous scientific standards, two distinct interpretations of this framework must be explicitly separated:

1. **Heuristic Claim:** Reality can be modeled *as if* it were a framestacked rendering pipeline. This serves as a powerful pedagogical tool and generates structural analogies across physics and systems architecture.
2. **Ontological Claim:** Reality is *literally* a computational rendering process executing on a lower-dimensional boundary. This is a speculative metaphysical research program.

The paper primarily defends the heuristic claim while framing the ontological position as a candidate research program to be constrained by empirical observation.

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1.2 Consciousness Telemetry Pipeline & Stacked History Stack

Visual information propagates at the speed of light c , whereas acoustic wavefronts travel through atmosphere at ≈ 343 m/s. Furthermore, biological neural transduction imposes a processing delay of tens to hundreds of milliseconds.

Conscious experience (“Now”) does not perceive raw, instantaneous state updates. Instead, the brain functions as a **telemetry buffering pipeline**: it receives asynchronous, delayed data packets, applies predictive processing, and compiles a temporally synchronized state into a *Stacked History Stack*. Conscious perception is structurally an archaeological reconstruction of historical event logs.

2 Formal Core Model

Let $\mathcal{B}$ represent the Hilbert state space of a lower-dimensional (two-dimensional) boundary manifold. The evolution of the universe is modeled as a discrete state update loop—termed *framestack execution tick* $n \in \mathbb{Z}$:

$$B_{n+1} = F(B_n), \quad B_n \in \mathcal{B} \quad (1)$$

where $F : \mathcal{B} \rightarrow \mathcal{B}$ is the discrete state transition operator governing boundary dynamics.

To account for the emergence of macroscopic three-dimensional spacetime and matter fields, we introduce a phenomenological projection operator Π . The mapping $\Pi(B_n)$ projects two-dimensional boundary state configurations onto emergent three-dimensional bulk fields:

$$\Pi(B_n) = (g_{\mu\nu}, \Psi, A_\mu, \dots) \quad (2)$$

where $g_{\mu\nu}$ represents the metric tensor of 3D bulk spacetime, Ψ represents matter fields, and A_μ represents gauge field connections.

The projection operator Π is formalized via a Tensor Network Contraction mapping, specifically utilizing the Multiscale Entanglement Renormalization Ansatz (MERA). MERA layer-by-layer coarse-graining utilizes local disentanglers u and isometries w with bond dimension χ , bounding boundary entanglement entropy via $S \leq \ln \chi$. While Π currently serves as a phenomenological mapping, future formalization must specify the exact isometric tensor constraints required to guarantee emergent Lorentzian bulk geometry.

Observables are defined as functionals $O[\Pi(B_n)]$. Causality is the partial ordering induced by the dependency directed acyclic graph (DAG) of F and Π :

$$i \prec j \iff O_j \text{ depends on } O_i \quad (3)$$

Rendering tax is defined as a cost functional over the bulk projection:

$$C[B_n] = \int_{\Sigma} \epsilon_{\text{bind}}(x) d^3x \quad (4)$$

Dimensionally, the cost functional $C[B_n]$ corresponds to local entanglement entropy S_{ent} and tensor-network bond-dimension density across the boundary projection interface. Mass m is interpreted as the localized resistance to changing $\Pi(B_n)$, yielding the accounting identity:

$$m \sim \frac{1}{c^2} \frac{\delta C}{\delta \Pi} \quad (5)$$

This does not replace $E = mc^2$; it reframes the mass-energy equivalence relation as an accounting identity for projection overhead.

3 Related Work & Literature Engagement

This framework bridges several established research programs in theoretical physics and computer science:

- **Holographic Principle:** 't Hooft [1], Susskind [2], and Maldacena [3] established that 3D gravitational bulk dynamics can be dual to 2D boundary quantum field theories.
- **Emergent Gravity:** Jacobson [4] derived Einstein's field equations from thermodynamics, and Verlinde [5] modeled gravity as an entropic force.
- **Digital Physics:** Zuse [9], Fredkin [10], and Wolfram [11] explored cellular automata as foundational models for physical law.
- **It from Bit:** Wheeler [7] proposed that physical entities derive their existence from binary information choices.
- **Computational Bounds:** Landauer [6] linked information erasure to physical entropy, and Bekenstein [12] bounded informational capacity by horizon area.
- **Predictive Processing:** Friston [8] formalized biological perception as an active inference telemetry pipeline.

4 Hardware Constraints & Defense of Lorentz Invariance

Physical constants [16] are interpreted as operational bounds of the execution substrate:

- **Planck Length:** $\ell_P \approx 1.6 \times 10^{-35}$ m — spatial resolution grid / pixel size limit.
- **Planck Time:** $t_P \approx 5.4 \times 10^{-44}$ s — clock cycle duration / framerate update interval.
- **Speed of Light:** $c = 3.0 \times 10^8$ m/s — maximum bus bandwidth / signal propagation limit.

4.1 Defense of Lorentz Invariance

A common objection to discrete lattice models is that spatial pixelation violates Lorentz invariance (special relativity). In this framework, lattice discretization at the Planck scale does not automatically break macroscopic Lorentz symmetry. Potential Lorentz-violating operators are suppressed at accessible energy scales by higher-dimensional Planck-scale energy cutoffs (E/M_{Planck}), preserving macroscopic Lorentz invariance.

4.2 Sub-Pixel Quantum Field Antialiasing

Continuity in macroscopic space is maintained through quantum field fluctuations that act as sub-pixel antialiasing. Mathematically, this is modeled as an operator-valued distribution smearing across test functions $f(x)$:

$$\hat{\phi}(f) = \int \hat{\phi}(x) f(x) d^4x \tag{6}$$

smoothing spatial discretization across neighboring Planck cells.

5 Computational Overhead of 3D Mass & Mass Taxonomy

Standard quantum chromodynamics (QCD) demonstrates that bare valence quark rest masses account for only $\approx 1\%$ of nucleon rest mass (protons and neutrons). The remaining $\approx 99\%$ originates predominantly from QCD gluon binding energy [13, 14, 15].

In our framework, 3D rest mass represents the **rendering tax** required to sustain localized 3D spatial field distortions against the 2D boundary plane. We delineate two operational mass resource classes:

1. **Dynamic Runtime Update Tax (Hadronic Mass):** Composite hadrons incur rest mass primarily through non-perturbative QCD gluon binding energy, representing active update overhead on the boundary.
2. **Static Registration Overhead (Leptonic Mass):** Free elementary particles (e.g., electrons) generate rest mass through direct Yukawa coupling to the Higgs field, acting as static state registration.

6 Quark Confinement as Runtime Exception Handling

Quarks carry color charge and cannot exist as isolated free particles [13, 15]. The potential energy between a quark-antiquark pair grows linearly with distance due to non-perturbative QCD gauge field dynamics:

$$V(r) \sim \sigma r, \quad \sigma \approx 1 \text{ GeV/fm} \quad (7)$$

When energy is injected to separate a quark pair, the gluon flux tube snaps, instantiating a new quark-antiquark pair via $E = mc^2$ (hadronization). As a pedagogical systems-engineering analogy, quark confinement functions as an automated **runtime exception handler** (catch-block) that maintains system data integrity by preventing unconfined color charge segmentation faults.

7 Extended Systems Engineering Mappings

7.1 Quantum Entanglement as Correlated Boundary References

Entangled particles share **correlated boundary-state references** on the 2D holographic boundary plane. Spatially separated 3D coordinates reflect common boundary memory addresses, naturally satisfying the No-Signaling theorem without implying classical data transfer.

7.2 Wavefunction Collapse as Environmental Decoherence

Measurement does not require a conscious observer. Wavefunction collapse represents environmental decoherence, wherein unobserved states execute lazy evaluation until state interaction forces procedural rasterization.

7.3 2D Biological Source Code

Linear DNA strings (ACGT) function as 2D source code compiled by ribosomal machinery into 3D structural protein engines driven by electron flow.

8 Falsifiability and Experimental Test Beds

To qualify as a physical theory, the model must yield testable deviations from standard physics:

1. **Planck-Scale Dispersion:** High-energy cosmic photons may exhibit energy-dependent arrival time delays due to Planck-grid discretization.
2. **Holographic Noise:** High-precision laser interferometers may detect spacetime jitter induced by boundary quantization.
3. **Sub-Millimeter Gravity Deviations:** Short-distance gravitational tests may detect corrections to Newton’s law predicted by emergent gravity models.
4. **Complexity Bounds:** Quantum information systems may reveal complexity-theoretic limits on multi-party entanglement scaling.

9 Master Systems Comparison Matrix

Table 1 summarizes the structural mappings developed throughout this manuscript, pairing each physical concept with its standard-physics caveat, its systems-engineering analogy, and its explicit epistemic status within this research program.

Table 1: System Architecture Mapping of Physical Concepts.

Physical Concept	Con-	Standard Physics Caveat	Systems Analogy	Epistemic Status	Sta-
Planck length ℓ_P		QG scale; not a proven pixel	Spatial resolution grid	Speculative	
Planck time t_P		QG scale; not a proven tick	Framerate duration	Speculative	
Speed of light c		Lorentz invariant; max causal speed	Bus bandwidth / latency cap	Heuristic	
Quantum Fields		Continuum QFT; UV cutoff needed	Sub-pixel antialiasing layer	Strong Analogy	
Holographic Boundary		Proven in AdS/CFT; open in dS	Frame buffer / 2D source array	Plausible	
Nucleon Mass		$\approx 99\%$ non-perturbative QCD	Dynamic runtime update tax	Strong Analogy	
Electron Mass		Higgs Yukawa coupling	Static registration overhead	Strong Analogy	
QCD String Tension σ		$V(r) \sim \sigma r$, $\sigma \approx 1$ GeV/fm	Data-line deformation cost	Strong Analogy	
Quark Confinement		Non-perturbative gauge property	Encapsulation / integrity guard	Strong Analogy	
Hadronization		Vacuum pair creation via $E = mc^2$	Exception handler + instantiation	Strong Analogy	
Entanglement		No-signaling correlations only	Correlated boundary references	Plausible	
Wavefunction Collapse		Environmentally induced decoherence	State resolution / lazy eval	Heuristic	
2D DNA $\rightarrow$ 3D Protein		Central dogma; biochemical folding	Source code $\rightarrow$ compiled engine	Pedagogical	
Sensory Latency		Finite neural/signal propagation	Buffered telemetry pipeline	Plausible	
Conscious “Now”		Predictive processing reconstruction	Stacked history stack	Plausible	

10 Conclusion & Open Problems

10.1 Preemptive Epistemic Disclaimer

No direct derivation of Einstein’s field equations or the Standard Model from the framestack update rule is currently claimed.

10.2 Open Problems

Future mathematical development must focus specifically on:

1. Constraining the isometric tensor properties of the projection operator Π .
2. Deriving the functional form of Einstein’s field equations,

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (8)$$

directly from MERA entanglement entropy variation under state updates.

Pending quantitative experimental validation, the Emergent Spacetime Rendering Engine functions as an integrative, heuristic research program bridging quantum field theory, holography, and systems design.

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